SPOKEN ARABIC DIGIT CLASSIFICATION USING RNN, LSTM AND GRU ON MFCC SEQUENCES

Deep Learning — Assignment 4, Task 2

Group 2

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1. Introduction

This work studies sequence classification with recurrent neural networks. The task is to recognise a spoken Arabic digit (0–9) from a variable-length sequence of Mel-Frequency Cepstral Coefficient (MFCC) vectors. Three recurrent cell types are compared: the vanilla RNN, the Long Short-Term Memory (LSTM) network and the Gated Recurrent Unit (GRU).

The networks are implemented in PyTorch. The number of recurrent layers, the number of hidden units and the cell type are tuned systematically. Training is stopped by a convergence criterion on the average training error rather than by a fixed number of epochs. As a bonus, bidirectional recurrent layers and temporal attention pooling are also tested.

2. Objectives

1. Build RNN, LSTM and GRU classifiers for variable-length MFCC sequences, using padding and packing.
2. Tune the number of recurrent layers, the number of hidden units and the cell type.
3. Stop training when $|\bar{L}^{(k)} - \bar{L}^{(k-1)}| < 10^{-4}$.
4. Report training error versus epochs, training and test accuracy, and the confusion matrix of the best model.
5. Visualise the MFCC sequences of the ten spoken digit classes.
6. Compare convergence behaviour and gradient flow across the three cell types.
7. (Bonus) Test GRU, bidirectional networks and temporal attention pooling.

3. Dataset and Preprocessing

The Spoken Arabic Digit dataset (UCI) provides pre-extracted MFCC sequences. Each utterance is a block of frames, and each frame is a 13-dimensional MFCC vector $x_t \in \mathbb{R}^{13}$. The utterance length varies from 4 to 93 frames (mean 39.9).

Table 1: Dataset summary.

Item	Value
Classes	10 (digits 0–9)
Original training file	6600 utterances (660 per class)
Original test file	2200 utterances (220 per class)
Feature dimension per frame	13 MFCC coefficients
Sequence length	4 to 93 frames (mean 39.9)

Splits. The original training file is divided 90%/10% (stratified, seed 42) into 5940 training and 660 validation utterances. The original test file (2200 utterances) is kept as the held-out test set. The best configuration is chosen using validation accuracy only.

Normalisation. Every MFCC coefficient is standardised with the mean and standard deviation computed from the 5940 training utterances only, and the same values are applied to the validation and test sets.

Batching. Utterances in a mini-batch are zero-padded to the longest sequence of that batch and passed to the network with `pack_padded_sequence`, so the padded frames never influence the hidden states.

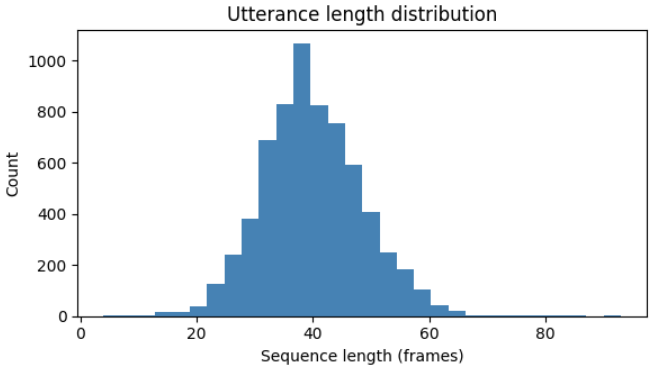


Figure 1: Distribution of utterance lengths in the training file.

4. Methodology

4.1 Network Structure

Each model has the form

$$\text{MFCC sequence} \longrightarrow \text{recurrent layers} \longrightarrow \text{sequence feature} \longrightarrow \text{linear layer (10 outputs)}.$$

For the standard models, the sequence feature is the final hidden state of the top layer (at the true last frame of each utterance). Dropout 0.2 is applied between recurrent layers when more than one layer is used.

4.2 Recurrent Cells

Vanilla RNN (tanh):

$$h_t = \tanh(W_{xh}x_t + W_{hh}h_{t-1} + b)\,.$$

LSTM:

$$
\begin{aligned}
i_t &= \sigma(W_i x_t + U_i h_{t-1} + b_i), & f_t &= \sigma(W_f x_t + U_f h_{t-1} + b_f),\\
o_t &= \sigma(W_o x_t + U_o h_{t-1} + b_o), & g_t &= \tanh(W_g x_t + U_g h_{t-1} + b_g),\\
c_t &= f_t \odot c_{t-1} + i_t \odot g_t, & h_t &= o_t \odot \tanh(c_t).
\end{aligned}
$$

GRU:

$$
\begin{aligned}
z_t &= \sigma(W_z x_t + U_z h_{t-1} + b_z), & r_t &= \sigma(W_r x_t + U_r h_{t-1} + b_r),\\
n_t &= \tanh\big(W_n x_t + r_t \odot (U_n h_{t-1}) + b_n\big), & h_t &= (1 - z_t) \odot n_t + z_t \odot h_{t-1}.
\end{aligned}
$$

The additive cell-state update of the LSTM and the gated interpolation of the GRU give the gradient a more direct path through time than the repeated tanh of the vanilla RNN.

4.3 Bonus Variants

Bidirectional: a second recurrence reads the sequence backwards and the two final states are concatenated.

Temporal attention pooling: instead of the last state, all valid hidden states h_t are combined,

$$
e_t = w^\top h_t + b, \qquad \alpha_t = \frac{\exp(e_t)}{\sum_{s=1}^{T} \exp(e_s)}, \qquad c = \sum_{t=1}^{T} \alpha_t h_t,
$$

where padded frames are masked out of the softmax. The vector c is passed to the linear classifier.

4.4 Loss Function

Cross-entropy loss is used. For epoch k, the average training error $\bar{L}^{(k)}$ is the mean loss over all training utterances, accumulated during the epoch.

4.5 Convergence Criterion

Training is stopped as soon as

$$
\left| \bar{L}^{(k)} - \bar{L}^{(k-1)} \right| < 10^{-4}.
$$

Two safeguards were added: the test is applied only from epoch 5 onward (`MIN_EPOCHS`=5), and training is capped at 100 epochs (`MAX_EPOCHS`=100). A run that reaches the cap without satisfying the criterion is marked "not converged" in the tables.

5. Experimental Setup

Table 2: Experimental configuration.

Item	Value
Recurrent cell	Vanilla RNN, LSTM, GRU
Recurrent layers	1, 2
Hidden units per layer	32, 64, 128
Number of baseline configurations	$3 \times 2 \times 3 = 18$
Optimizer	Adam, learning rate 10^{-3}
Batch size	64
Gradient clipping	global norm 5
Dropout (between layers)	0.2
Stopping rule	$\lvert\bar{L}^{(k)} - \bar{L}^{(k-1)}\rvert < 10^{-4}$ (from epoch 5, max 100)
Random seed	42
Hardware	CPU (Google Colab)

For the bonus study, the best baseline configuration of each cell type (chosen by validation accuracy) is retrained with: bidirectional, attention pooling, and both together.

6. Experimental Results

6.1 MFCC Feature Visualisation

Figure 2 shows one MFCC heatmap per spoken digit class (rows: coefficient index, columns: frame). The low-order coefficients (0–3) show the strongest time-varying structure, and the high-order coefficients look noisy. Figure 3 shows several utterances of the same digit: the pattern is similar, but the length and timing differ from speaker to speaker.

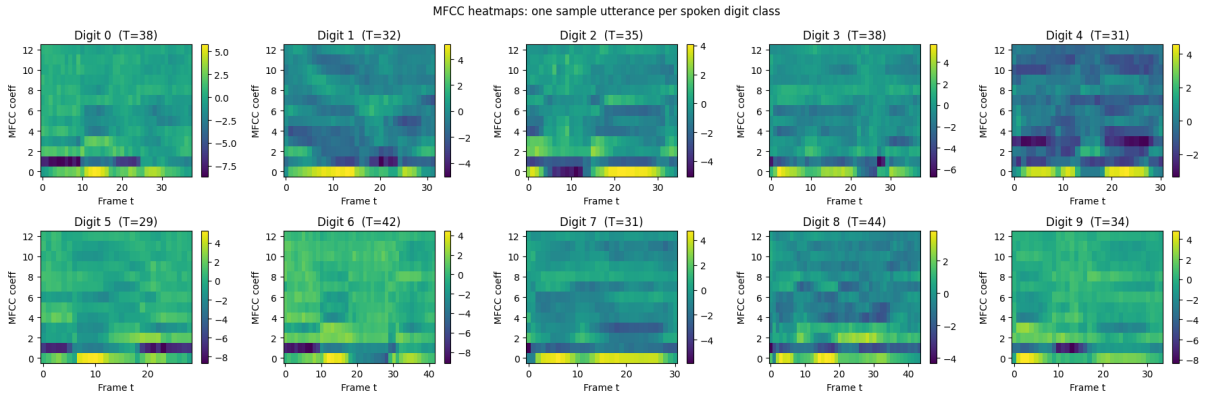


Figure 2: MFCC heatmaps, one sample utterance for each of the ten digit classes.

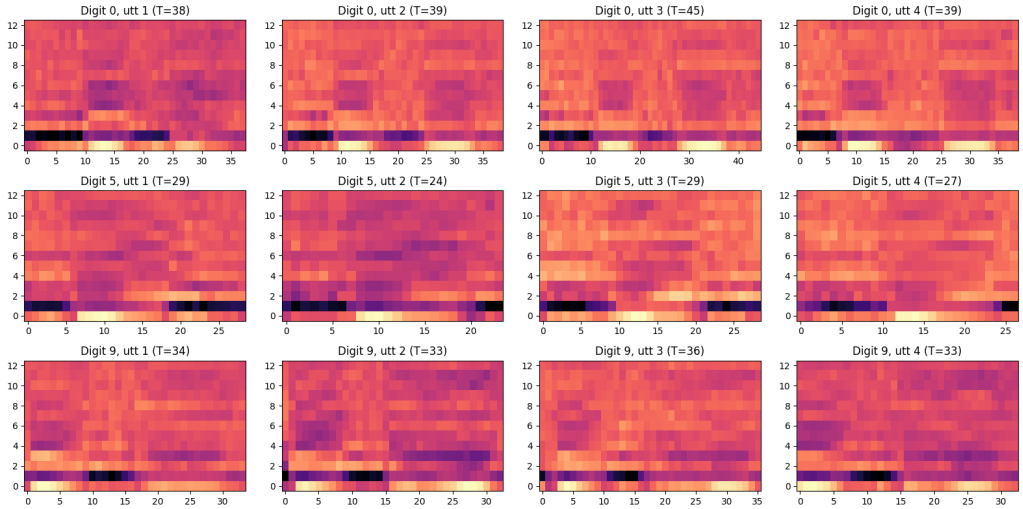


Figure 3: Within-class variability: four utterances each of digits 0, 5 and 9.

6.2 Average Training Error versus Epochs

Figure 4 shows the average training error for all 18 baseline configurations, with one panel for each cell type. Figure 5 shows the same curves on a logarithmic scale.

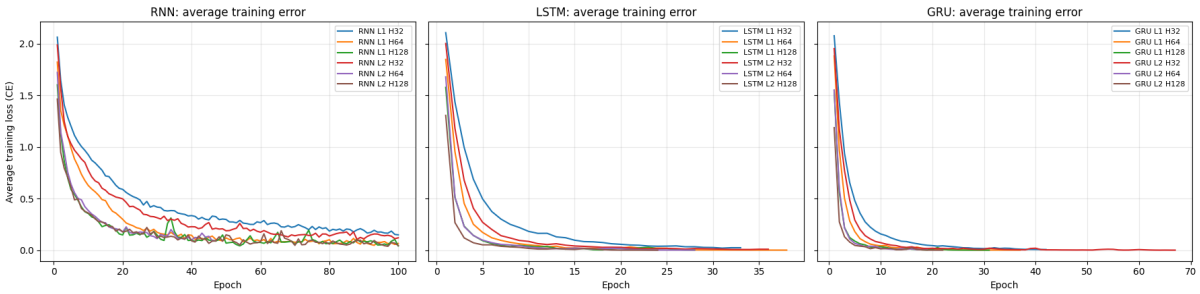


Figure 4: Average training error (cross-entropy) versus epochs for all hyperparameter sets.

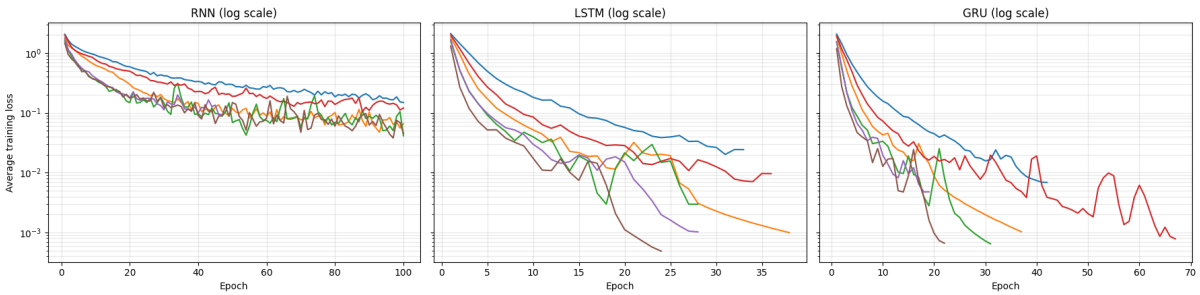


Figure 5: The same curves on a logarithmic scale.

6.3 Summary of All Baseline Runs

Table 3: Baseline hyperparameter search. Accuracies are in %. "Conv." means the 10^{-4} criterion was met before the 100-epoch cap.

Cell	Layers	Hidden	Epochs	Conv.	Final loss	Train	Val	Test
RNN	1	32	100	No	0.14881	96.06	92.42	85.36
RNN	1	64	100	No	0.06574	98.15	95.30	89.09
RNN	1	128	100	No	0.04140	99.48	98.48	94.64
RNN	2	32	100	No	0.12024	98.37	96.67	90.36
RNN	2	64	47	Yes	0.09282	97.32	95.91	93.05
RNN	2	128	100	No	0.04545	98.84	96.97	93.32
LSTM	1	32	33	Yes	0.02427	99.66	98.03	94.32
LSTM	1	64	38	Yes	0.00099	100.00	99.09	95.45
LSTM	1	128	28	Yes	0.00296	99.92	99.09	95.95
LSTM	2	32	36	Yes	0.00963	99.97	98.79	94.64
LSTM	2	64	28	Yes	0.00102	100.00	99.39	95.41
LSTM	2	128	24	Yes	0.00049	100.00	99.24	95.45
GRU	1	32	42	Yes	0.00688	99.88	98.64	94.27
GRU	1	64	37	Yes	0.00103	100.00	99.39	96.09
GRU	1	128	31	Yes	0.00065	100.00	99.24	97.00
GRU	2	32	67	Yes	0.00078	100.00	99.55	96.27
GRU	2	64	19	Yes	0.00478	99.83	99.09	94.73
GRU	2	128	22	Yes	0.00066	100.00	99.09	96.41

Table 4: Mean over the six configurations of each cell type.

Cell	Mean train acc (%)	Mean test acc (%)	Mean epochs
Vanilla RNN	98.04	90.97	91.17
LSTM	99.92	95.20	31.17
GRU	99.95	95.80	36.33

The best configuration of each cell type, selected by validation accuracy, is:

Table 5: Best baseline configuration of each cell type.

Cell	Best config	Epochs	Train (%)	Val (%)	Test (%)	Time (s)
RNN	1 layer, 128 units	100	99.48	98.48	94.64	301.5
LSTM	2 layers, 64 units	28	100.00	99.39	95.41	358.5
GRU	2 layers, 32 units	67	100.00	99.55	96.27	454.0

6.4 Bonus: Bidirectional and Attention Models

Table 6: Bonus variants of the best configuration of each cell type. Accuracies are in %.

Cell	Layers	Hidden	Variant	Epochs	Final loss	Train	Val	Test
RNN	1	128	BiDir	32	0.02077	99.61	98.48	96.45
RNN	1	128	Attention	60	0.00070	100.00	99.09	97.00
RNN	1	128	BiDir+Attn	40	0.00086	100.00	99.39	97.14
LSTM	2	64	BiDir	16	0.00068	100.00	99.39	96.14
LSTM	2	64	Attention	37	0.00061	100.00	99.39	97.14
LSTM	2	64	BiDir+Attn	21	0.01139	99.87	99.39	96.36
GRU	2	32	BiDir	29	0.00100	100.00	99.70	97.55
GRU	2	32	Attention	44	0.00192	100.00	99.39	96.95
GRU	2	32	BiDir+Attn	26	0.00129	100.00	99.55	98.00

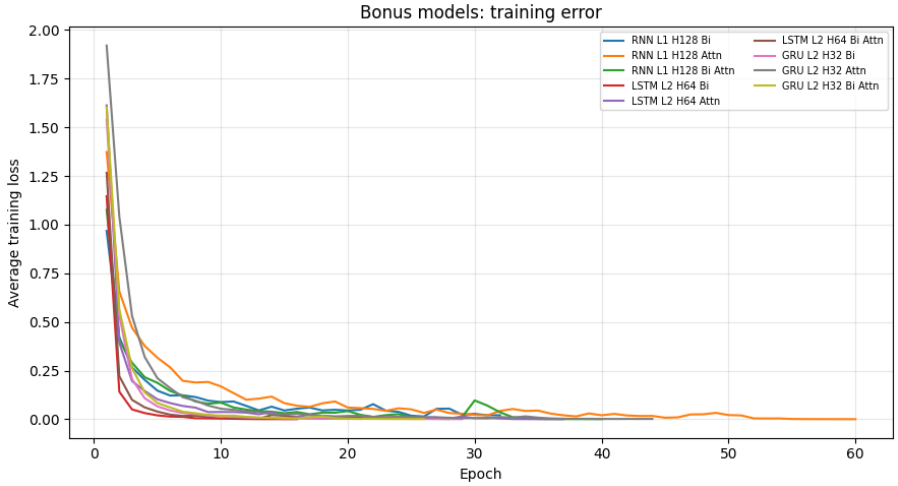


Figure 6: Average training error of the bonus models.

6.5 Best Models: Detailed Performance

Selecting by validation accuracy, the best standard model is the **2-layer GRU with 32 hidden units**. Over all runs including the bonus models, the best is the **2-layer bidirectional GRU with 32 hidden units**.

Table 7: Final accuracy of the two selected models.

Model	Epochs	Train (%)	Val (%)	Test (%)
GRU, 2 layers, 32 units (best standard)	67	100.00	99.55	96.27
BiGRU, 2 layers, 32 units (best overall)	29	100.00	99.70	97.55

Table 8: Per-class test metrics (2200 test utterances, 220 per class).

	GRU L2 H32			BiGRU L2 H32		
Digit	Precision	Recall	F1	Precision	Recall	F1
0	0.9142	0.9682	0.9404	0.9188	0.9773	0.9471
1	0.9690	0.9955	0.9821	0.9821	1.0000	0.9910
2	0.9641	0.9773	0.9707	0.9954	0.9864	0.9909
3	0.9577	0.9273	0.9423	0.9718	0.9409	0.9561
4	0.9729	0.9773	0.9751	0.9729	0.9773	0.9751
5	0.9234	0.9864	0.9538	0.9778	1.0000	0.9888
6	0.9854	0.9182	0.9506	1.0000	0.9409	0.9696
7	0.9631	0.9500	0.9565	0.9861	0.9682	0.9771
8	0.9904	0.9364	0.9626	0.9771	0.9682	0.9726
9	0.9954	0.9909	0.9932	0.9777	0.9955	0.9865
Macro avg	0.9636	0.9627	0.9627	0.9760	0.9755	0.9755

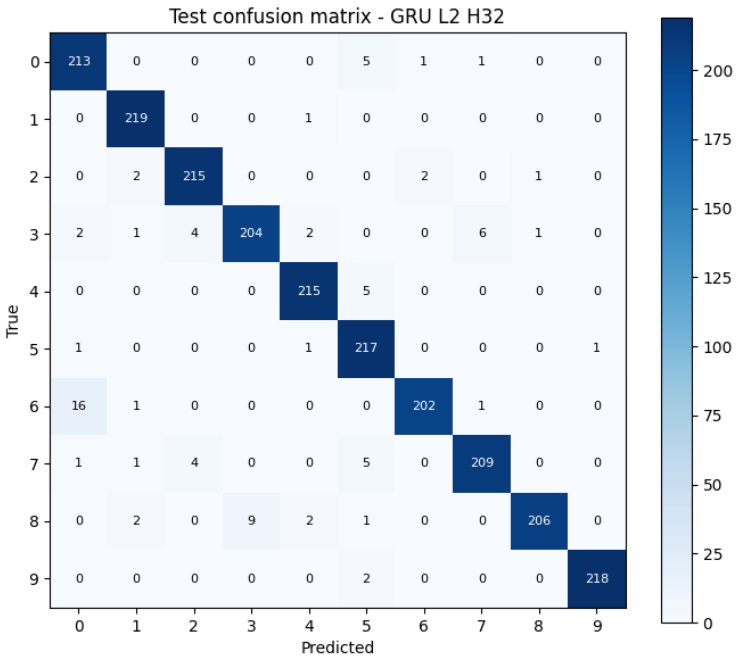


Figure 7: Test confusion matrix of the best standard model (GRU, 2 layers, 32 units); test accuracy 96.27%.

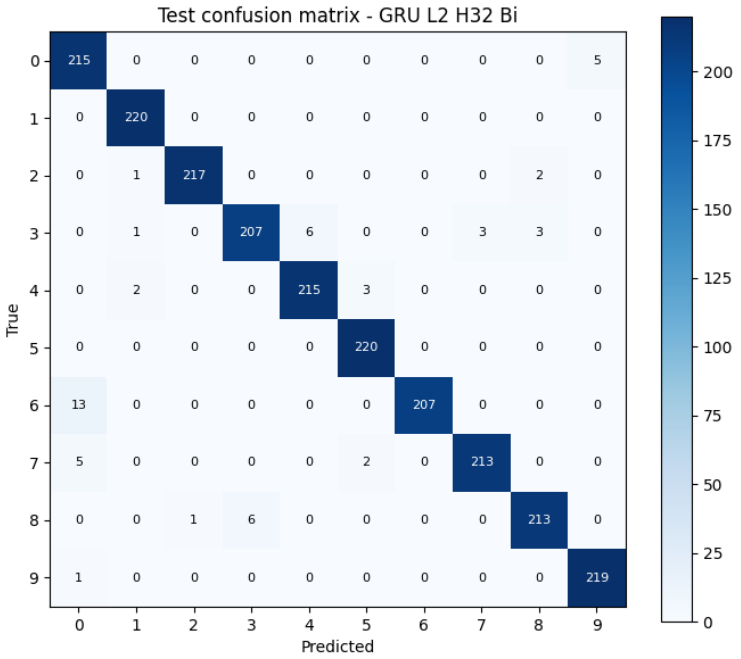


Figure 8: Test confusion matrix of the best overall model (BiGRU, 2 layers, 32 units); test accuracy 97.55%.

In both matrices, the most frequent error is a spoken "6" recognised as "0" (16 cases for the standard GRU, 13 for the BiGRU). The next most common is "8" recognised as "3" (9 and 6 cases). The bidirectional model reduces the number of wrong predictions of digit 5 from 16 to 5 and the total number of errors from 82 to 54 out of 2200.

7. Gradient Flow Analysis

Two quantities were recorded for the best configuration of each cell type.

(a) Gradient norm during training. The global gradient norm was measured before clipping and averaged over each epoch (Figure 9).

(b) Gradient through time. For 512 test utterances with at least 30 frames, the norm $\|\partial L/\partial x_t\|$ was computed for each of the last 30 frames and plotted against the distance from the last frame (Figure 10). This shows how strongly a frame far from the output still influences the loss.

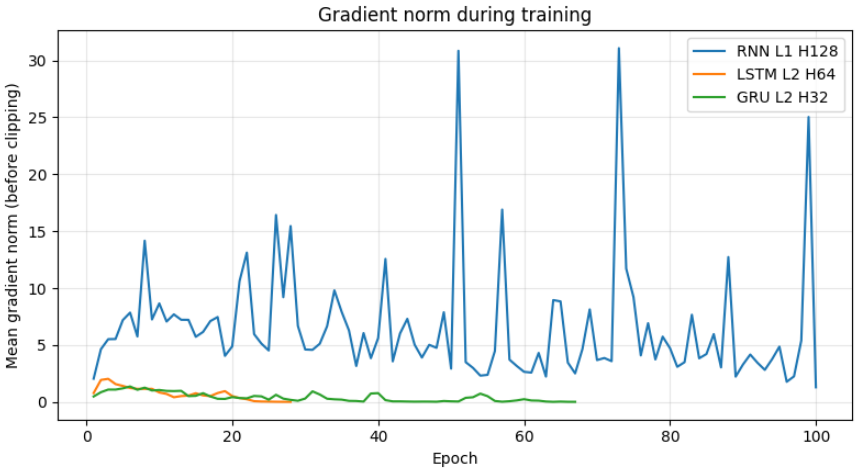


Figure 9: Average gradient norm (before clipping) per epoch.

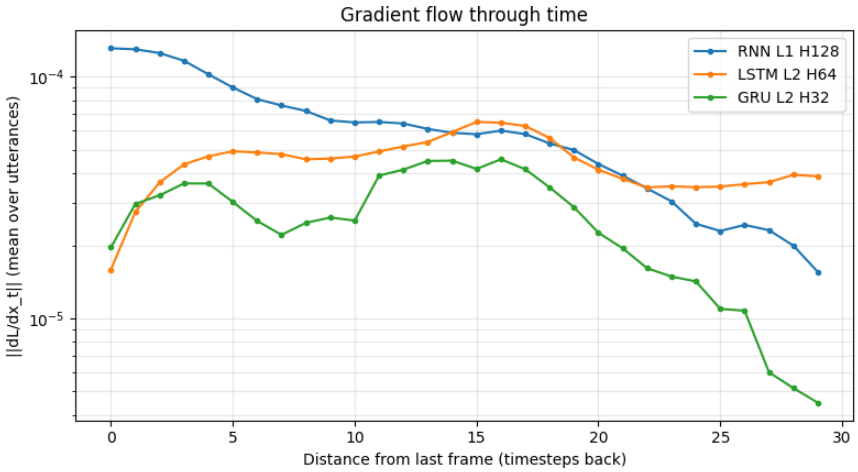


Figure 10: Gradient of the loss with respect to the input frame versus distance from the last frame (log scale).

Observations.

- The vanilla RNN has an irregular gradient norm with repeated spikes of about 25–31 (for example near epochs 51, 73 and 99), which is far above the clipping threshold of 5. The LSTM and the GRU have small and smooth gradient norms (mostly below about 2) that fall towards zero as the loss falls.
- In the through-time plot, the RNN gradient decreases steadily from about 1.3×10^{-4} at the last frame to about 1.5×10^{-5} thirty frames earlier, a drop of roughly 8×. This is consistent with gradient decay through repeated tanh multiplications.
- The LSTM curve stays within a narrow band (about 1.6×10^{-5} to 6.5×10^{-5}) and is not lower at distance 29 than at distance 0. The GRU curve falls by about 4× over the same range, but not monotonically.

Caution. This probe is applied to already trained models, and a model with near-zero loss produces small gradients overall. The absolute levels of the three curves therefore cannot be compared directly, and only their shapes are informative. It is also a single run with one seed.

8. Discussion

Accuracy. Gated cells clearly outperform the vanilla RNN. On average, the LSTM and the GRU reach 95.20% and 95.80% test accuracy, against 90.97% for the RNN. The best RNN (1 layer, 128 units) reaches 94.64%, about 1.6 points below the best GRU (96.27%) and about 0.8 points below the best LSTM (95.41% at the validation-selected configuration, 95.95% for 1 layer and 128 units).

Convergence. All twelve LSTM and GRU runs satisfied the 10^{-4} criterion, after 19 to 67 epochs. Only one of the six RNN runs (2 layers, 64 units) did. The other five reached the 100-epoch cap because the RNN training loss keeps fluctuating by more than 10^{-4} from one epoch to the next. The one RNN run that did stop early did so at a relatively high loss (0.093) with 97.32% training accuracy, lower than four of the other five RNN runs, which shows that the criterion can stop a noisy run before it has trained well.

Effect of size. For the RNN, a larger hidden layer helps (test accuracy 85.36%, 89.09% and 94.64% for 32, 64 and 128 units with one layer). For the LSTM and the GRU, the differences between configurations are small, and the ranking by test accuracy does not follow the ranking by validation accuracy. Depth did not give a consistent gain: for example, the 1-layer GRU with 128 units has the highest baseline test accuracy (97.00%), but the 2-layer GRU with 32 units has the highest validation accuracy and is therefore the selected model.

Bonus variants. Bidirectional reading and attention pooling improved most models. The largest gain is for the vanilla RNN, from 94.64% to 97.00% with attention and 97.14% with both. The LSTM improved from 95.41% to 97.14% with attention. The GRU improved from 96.27% to 97.55% with bidirectional reading and to 98.00% with both. Because a single utterance is $1/2200 \approx 0.045$ points, differences of a few tenths of a point (for example between 97.14% and 97.55%) correspond to only a handful of utterances and may be due to chance; only one seed was run for each configuration.

Validation versus test. Validation accuracy (99–99.7%) is clearly higher than test accuracy (95–98%). The validation set is a random part of the original training file, so it probably contains speakers that also appear in the training part, while the test file is a separate set of utterances. The test accuracy is therefore the more realistic estimate for new data, and speaker-wise validation splits would give a more reliable model selection.

9. Conclusion

RNN, LSTM and GRU classifiers were trained on MFCC sequences of spoken Arabic digits with padded and packed batches and with the stopping rule $|\bar{L}^{(k)} - \bar{L}^{(k-1)}| < 10^{-4}$. Across 18 baseline configurations, the gated cells (LSTM and GRU) trained faster, converged reliably and reached higher test accuracy than the vanilla RNN, whose training loss and gradient norm were noisy.

The best baseline model, a 2-layer GRU with 32 units, reached 100.00% training and 96.27% test accuracy. The best overall model, a 2-layer bidirectional GRU with 32 units, reached 100.00% training and 97.55% test accuracy. Adding attention to the bidirectional GRU gave 98.00% test accuracy, but its validation accuracy (99.55%) was lower than that of the bidirectional GRU without attention (99.70%), so it was not selected.

10. Key Experimental Values

Table 9: Summary of experimental values.

Item	Value
Dataset	Spoken Arabic Digit (MFCC sequences)
Train / validation / test utterances	5940 / 660 / 2200
Input per frame	13 MFCC coefficients (standardised)
Cells compared	Vanilla RNN, LSTM, GRU
Layers, hidden units	$\{1, 2\} \times \{32, 64, 128\}$
Loss	Cross-entropy
Stopping rule	$\lvert\bar{L}^{(k)} - \bar{L}^{(k-1)}\rvert < 10^{-4}$
Best standard model	GRU, 2 layers, 32 units
train / test accuracy	100.00% / 96.27%
Best overall model	BiGRU, 2 layers, 32 units
train / test accuracy	100.00% / 97.55%